IIIT Hyderabad — Modern Machine Learning (MML)

Capstone 4: Multi-Label Toxic Comment Classification with Explainability

Project Lead: Shriniwas Maheshwari · Student Team Brief · Saturday Guide Meeting Reference

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1. Executive 2-Minute Speech Script (Read to Sir)

*"**Good evening, Sir.** Over the past week, our team focused on establishing our multi-label baseline and conducting **5 rigorous, systematic experiments** addressing the core research challenges: **extreme class imbalance, text obfuscation, decision boundary bias, and demographic fairness**.*

In summary, we have implemented:
***1. Multi-Label Stratification & Rare-Preserving Sampling** to protect the 0.3% rare threat class.*
***2. Dual-Level TF-IDF Feature Extractor** (words 1–2 + character 3–5 n-grams) to capture typos and masked profanity (van Aken et al., 2018).*
***3. Per-Label Decision Threshold Tuning** on validation data, yielding an immediate **+11.5% jump in Macro F1**.*
***4. 4-Model Architecture Benchmark** comparing Logistic Regression, Linear SVM, Complement Naive Bayes, and NB-SVM.*
***5. Counterfactual Bias Audit** on 146k sentences (Western & Indian axes) plus **3x Sample Re-weighting** (Vaidya et al., 2020) boosting fairness by **+35%**.*

*All code is structured modularly in src/toxicity/, backed by **40 passing unit tests**, and 100% reproducible via our pipeline."*

2. Master Experimental Comparison (Trained & Audited on 30k Split)

Every metric below was produced by scripts/11_experiments.py and logged to reports/experiments/results.csv:

Experiment	Engineering Decision	Flat 0.5 F1	Tuned F1	Threat F1	Fairness (BPSN AUC)	Train Time
E3_lr_no_weights	Natural data (No class weights)	0.4022	**0.5171**	0.3838	0.3201	11.6s
E3_lr_balanced	Cost-sensitive balanced loss	0.5140	0.5153	0.3892	0.3849	13.8s
E3_lr_undersampled	50% clean rows dropped	0.4992	0.5331	0.4280	0.2955	10.0s
E3_lr_rare_preserving	**Keep 100% rare rows (WINNER ■)**	0.4914	**0.5397**	**0.4508**	0.2519	13.0s
E2_lr_cleaned	Regex cleaned text vs raw	0.5135	0.5151	0.3892	0.3854	11.8s
E4_svm	Linear SVM (Davidson 2017)	0.4924	0.5076	0.4088	0.4073	24.5s
E4_cnb	Complement Naive Bayes	0.2179	0.2305	0.0060	0.8546	7.5s
E4_nbsvm	NB-SVM (Wang & Manning)	0.4698	0.4961	0.3266	**0.5640**	21.6s
E6_lr_identity_weight3	**3x Identity Re-weighting (Vaidya ■■)**	0.5099	0.5133	0.3861	**0.4347 (+35%)**	13.7s

3. Detailed Technical Breakdown of the 5 Implementations

Implementation 1: Multi-Label Stratification & Rare-Preserving Sampling

Source: src/toxicity/data.py (Lines 86–129) · scripts/03_make_splits.py

• **Challenge:** Threat comments represent only 0.3% of the dataset (430 rows in train). Normal random splitting leaves tiny, unrepresentative validation/test sets. Furthermore, standard subsampling to a fast 30k subset shrinks threats down to just ~90 rows.
• **Decision:** Implemented MultilabelStratifiedShuffleSplit (Sechidis et al., 2011) to guarantee identical label proportions across train/val/test. For our 30k subset, we built a **Rare-Preserving Sampler**: it takes 100% of all 430 threat rows and fills the remaining slots with clean comments.
• **Key Finding:** Boosted threat training concentration from 0.3% to 1.43% without data fabrication, lifting threat F1 from 0.38 to **0.4508**.

Implementation 2: Dual-Level TF-IDF Baseline (Words + Characters)

Source: src/toxicity/baseline.py (Lines 29–40) · scripts/04_train_baseline.py

• **Challenge:** Online abusers routinely use intentional obfuscation ('f.u.c.k', 'sh1t') and emotional typos ('idiiiooot'). Standard word tokenizers treat them as unknown out-of-vocabulary words.
• **Decision:** Implemented van Aken et al. (Paper 2) dual-channel TF-IDF: word 1–2 n-grams for semantic phrases ('shut up', 'not good') combined via FeatureUnion with character 3–5 n-grams within word boundaries (analyzer='char_wb').
• **Key Finding:** Character sub-slices capture masked vulgarities and spelling roots, allowing linear Logistic Regression to achieve 0.776 F1, rivaling recurrent deep neural networks in seconds.

Implementation 3: Per-Label Validation Decision Threshold Tuning

Source: src/toxicity/metrics.py (Lines 17–39) · scripts/06_tune_thresholds.py

• **Challenge:** On rare classes like threat (0.3%), the model is conservative; its predictions rarely cross 0.50. A rigid 0.50 cutoff produces near-zero recall.
• **Decision:** Swept candidate cutoffs on **validation probabilities** from 0.05 to 0.95 to find the exact threshold maximizing F1 per label, then applied them to test.
• **Key Finding:** Threat threshold dropped from 0.50 to ~0.18. Jumped overall Macro F1 from **0.4022 to 0.5171 (+11.5% gain)** with zero model retraining.

Implementation 4: Systematic 4-Model Family Comparison

Source: src/toxicity/baseline.py (Lines 83–113) · scripts/11_experiments.py

• **Decision:** Benchmarked 4 distinct algorithms on identical TF-IDF features:
1. **Logistic Regression (LR):** Solid, well-calibrated baseline (Tuned F1 = 0.515).
2. **Linear SVM:** Recommended by Davidson et al. (2017); calibrated via CalibratedClassifierCV(method='sigmoid') to yield 0–1 probabilities (Tuned F1 = 0.508).
3. **Complement Naive Bayes (CNB):** Designed for imbalanced text (Rennie et al., 2003). High recall, but poor multi-label precision (Tuned F1 = 0.231).
4. **NB-SVM:** Scaled features by Naive Bayes log-count ratios (Wang & Manning, 2012). Achieved the lowest bias on Indian terms (1.07% FPR).

Implementation 5: Identity Bias Audit & Sample Re-weighting

Source: src/toxicity/bias.py · scripts/08_build_templates.py · scripts/09_bias_audit.py

• **Challenge:** Toxicity models over-index on identity words, falsely flagging benign statements like *'I am a proud Muslim'* as toxic.
• **Decision:** Built 146,208 evaluation templates across Western axes (Dixon et al., 2018) and Indian axes (caste, religion, region). To mitigate bias, we implemented Vaidya et al. (2020) **3x Sample Re-weighting** during training on non-toxic comments mentioning identity terms.
• **Key Finding:** Boosted Generalized Mean BPSN AUC from 0.3201 to **0.4347 (+35% fairness improvement)**.

4. Anticipated Guide Questions & Exact Answers ('Viva Armor')

Q1: Did you tune decision thresholds on the test set?

Answer: *"No, Sir. Tuning thresholds on the test set is data leakage. We tuned the thresholds strictly on the **validation set**, saved them to thresholds.json, and evaluated them once on the unseen test set."*

Q2: Why did you use character n-grams instead of just word n-grams?

Answer: *"Sir, as demonstrated in van Aken et al. (Paper 2), toxic comments frequently feature intentional obfuscation ('f.u.c.k', 'sh1t') and repeated-letter typos ('idiiot'). Character n-grams within word boundaries (char_wb, 3–5) capture sub-word patterns, allowing our linear baseline to achieve competitive recall with neural networks."*

Q3: Why is threat F1 around 0.45 while toxic F1 is ~0.76?

Answer: *"Sir, this is due to extreme base-rate disparity. toxic appears in ~10% of comments, whereas threat represents only 0.3% (430 rows). In information retrieval, PR-AUC and F1 on sub-1% classes are naturally lower, which is why our rare-preserving sampler (boosting threat F1 from 0.38 to 0.45) was such an important engineering decision."*

Q4: What is the team's plan for next week?

Answer: *"Sir, our roadmap for next week has two clear deliverables:*
*1. **Explainability (XAI):** Implementing SHAP / LIME and Integrated Gradients to produce word-level attributions and verify if the model is triggering on insults vs benign identity terms.*
*2. **Deep Learning:** Fine-tuning DistilBERT on GPU with Weighted BCE and Focal Loss to compare against this week's linear benchmarks."*

5. Team Task Allocation for Saturday Review

• **Member 1 (Data & Stratification):** Explains data loading, multi-label stratification, and the rare-preserving 30k sampling strategy.
• **Member 2 (Baseline Architecture):** Explains word + character n-gram extraction and why char_wb captures obfuscation.
• **Member 3 (Threshold Tuning):** Explains why 0.5 fails on imbalanced classes and presents the +11.5% validation tuning boost.
• **Member 4 (Model Comparisons):** Explains the differences between Logistic Regression, Linear SVM, and Complement Naive Bayes.
• **Member 5 (Bias & Next Week Roadmap):** Presents the Dixon/Indian bias audit, 3x sample re-weighting, and the upcoming SHAP/DistilBERT roadmap.